Daniel A. Delgado

+1 (754) 837-7742 • danieldel@ufl.edu

EDUCATION

PhD in Computer Science
Degree Earned: *August 2025*
University of Florida, Gainesville, FL

- **Dissertation Title**: Enhancing Collaborator Communication through Shared-Gaze Visualizations in Augmented Reality

MS in Computer Science
Degree Earned: *May 2022*
University of Florida, Gainesville, FL

- Advisor: Jaime Ruiz, PhD (*Ruiz HCI Lab*)
- **Honors:** McKnight Doctoral Fellow, Graduate Student Preeminence Award

BS in Computer Engineering
Degree Earned: *August 2019*
Florida State University, Tallahassee, FL

- **Honors:** Magna Cum Laude

RESEARCH INTERESTS

Collocated Collaboration in Augmented Reality. Eye-Tracking. Shared-Gaze Visualizations. Human-Computer Interaction. User Experience Design. Human-Machine Interface. Human-Machine Collaboration. Bio-sensors. Qualitative and Quantitative Analysis.

PUBLICATIONS

Delgado, D. A. *Enhancing Collaborator Communication through Shared-Gaze Visualizations in Augmented Reality*. Doctoral Dissertation, University of Florida. 2025.

Delgado, D. A., Calvo, L. C., Bowers C. J., Ruiz, J. *When Is Self-Gaze Helpful? Examining Uni- vs Bi-directional Gaze Visualization in Collocated AR Tasks*. IEEE ISMAR. 2025.

Delgado, D. A., Barquero, A., Calvo, L. C., Wang, I., Ruiz, J., Anthony, L. *Navigating the Kitchen: Understanding User Needs for Task Guidance while Cooking.* ACM DIS. 2024.

Liu, Y., Park, J., **Delgado, D. A.**, Music, A., Berman, J., Ruiz, J., Kaber, D., Huang, H., and Zahabi, M. "Virtual Reality as a Platform for Upper-Limb Prosthetic Control Training and Early-Stage Design." Special Issue on User-Centered Innovations in Upper-Limb Prosthetics, (2025). (In Review)

Sherman, I., **Delgado, D. A.**, Gilbert, J. E., Ruiz, J., & Traynor, P. (2024). *"I used to live in Florida": Exploring the Effects of Spam Call Warning Accuracy on Callee Decision-Making.* USEC. 2024.

Junho, P. , Music A., **Delgado D. A.**, Liu, Y., Berman, J., Zahabi, M., Ruiz, J., Kaber, D., Dang, H. (2023). Cognitive Workload and Usability of Virtual Reality Simulation for Prosthesis Training. IEEE SMC. 2023.

Sherman, I., **Delgado, D. A.**, Gilbert, J. E., Ruiz, J., & Traynor, P. (2021, August). *Characterizing User Comprehension in the STIR/SHAKEN Anti-Robocall Standard*. In TPRC49: The 49th Research Conference on Communication, Information and Internet Policy.

POSTERS

Delgado, D.A, Ruiz, J. *Evaluation of Shared-Gaze Visualizations for Virtual Assembly Tasks.* IEEE VR 2024.

AI Days Postering Session, The University of Florida – Gainesville, Florida

AWARDS

Best Poster Honorable Mention: IEEE VR 2024
Spring 2024

TEACHING

Teaching Assistant Programming II: Algorithm Abstraction & Design

Spring 2025

- Grading exams and attending to discussion questions.

Instructor of Record Programming I: Python

Fall 2023

- Presenting class lectures, structuring lesson plans, and managing teaching assistants.

Teaching Assistant Programming II: Java

Summer 2023

- Hosting class labs, office hours, and guest lectures.

MENTORING

Undergraduate Student Mentor

Fall 2024 – Spring 2025

- Mentored an undergraduate student in research projects involving augmented reality and industrial applications.

RESEARCH AND PROJECTS

Graduate Research

Shared-Gaze Visualizations in Augmented Reality

August 2023 – Present

- Enhancing collaborator communication through shared-gaze visualizations in augmented reality for industrial tasks.

NSF EMG Human-Machine Interface

August 2021 – December 2023

- Classifying EMG signals for human-machine interface.

Adaptive Task Guidance for Colocated Collaborators

August 2021 – May 2023

- Designing a context aware adaptive system for synchronous collaborative tasks.

ENKix: Enabling Knowledgeable Task Guidance In the extremes

August 2021 – December 2022

- Designing a novel and effective AR-based task guidance system designed to handle the characteristics found at the extremes.

Multimodal Affect Detection System

June 2020 – August 2021

- Utilizing multi-dimensional Convolutional Neural Networks for audio-visual emotion recognition.

Characterizing User Comprehension in the STIR/SHAKEN

February 2021 – July 2021

- Analyzing and coding user surveys for a new anti-spam protocol.

Communicating with Computers

September 2019 – June 2021

- Evaluating a multi-modal interactive AI through participant studies.

Undergraduate Research and Projects

IEEE SoutheastCon Robotics Competition, *Florida State University*

- Designing and implementing computer vision devices on stand-alone robot for automatic location and object detection.

Innovatech Engineering, LLC, *Tallahassee, Florida*

- Designing and assembling custom PCB hardware for underwater scuba diving tank refill pump and underwater depth detector.

Electrolytic Battery Cell Synthesis, *Florida State University*

- Synthesizing and testing lithium-ion battery cells for efficiency and duration.

Gonzalez and Sons Engineering Contractor, *Medley, Florida*

- Designed safety tags for heavy duty construction equipment and performed on-sight evaluations.

PROFESSIONAL MEMBERSHIPS

Association of Computing Machinery (ACM)
January 2020 – Present

Institute of Electrical and Electronics Engineers (IEEE)
December 2018 – Present

Special Interest Group on Computer–Human Interaction (SIGCHI)
January 2020 – January 2021

Special Interest Group on Artificial Intelligence (SIGAI)
January 2020 – January 2021

SERVICE

Conference Reviewer **ACM CHI Case Studies 2025**
Conference Reviewer **2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)**
Conference Reviewer **ACM CHI Conference on Human Factors in Computing Systems 2023**
Conference Reviewer **ACM CHI Conference on Human Factors in Computing Systems 2022**

TALKS

ACM DIS 2024 – Copenhagen, Denmark
IEEE VR 2024 – Orlando, Florida
Human-Centered Computing International Summit 2019 – Universidad El Bosque
IEEE Southeast Conference 2019 – Huntsville, Alabama